Following the Compute Money: An On-Chain Forensic Analysis of GPU-Backed Credit and Decentralized Compute Markets

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Code and frozen data: https://github.com/uwecerron/covalent_onchain_compute

***Abstract*—Several synthetic-dollar protocols take stablecoin deposits and lend them to companies that operate GPU clusters. The GPUs, the borrowers and the loan contracts are off-chain, but the debt is recorded on a public chain. We rebuild one such protocol, USD.AI on Arbitrum, from 55,954 decoded events between 2025-05-12 and 2026-10-22, using the Covalent GoldRush API and event signatures taken from the protocol's source code. We find 39 loans to 13 borrowers, with $290.1 M originated and $182.2 M outstanding, rates between 7.0% and 15.0%, and no defaults. The largest borrower received 45.6% of all lending (HHI 0.273). The protocol's own hardware-token metadata puts about 4,560 current-generation GPUs behind the active loans at a 66.8% loan-to-value on a three-year life. The sUSDai vault grew 8.17% a year and 93.7% of that yield has arrived as cash, but more of the cash came from stablecoin issuers (M^0 and PYUSD) than from borrower interest. The median redemption waited 13.0 days, and redemptions do not move with GPU rental prices. Checked against the vault's own accounting at one block, our loan book is within 0.22% and our share count within 0.00%. The same check shows that 23.2% of reported assets is escrow for loans that do not exist on-chain. The escrowed funds were redeemed to PYUSD and sent to an off-ramp wallet. Over the same period, Akash and Livepeer, two established decentralized compute markets, took in $4.47 M, about 65 times less than the vault lent. The code and a hash-verified dataset reproduce every number.**

***Index Terms*—DeFi, stablecoins, GPU finance, on-chain forensics, real-world assets, decentralized compute, ERC-4626, ERC-7540**

I. Introduction

Several synthetic-dollar protocols now take stablecoin deposits on-chain and lend them to companies that operate GPU clusters, paying depositors from the interest. USD.AI is one of the larger ones. It has a $100 M credit facility from Bullish [1], and its tokens have been proposed as collateral on Aave [2]. Futures on GPU rental prices are also being built so lenders can hedge the same risk [3]. Lenders, risk curators and auditors are pricing GPU-backed credit, but nobody has published an independent account of what the chain shows about it.

The GPUs, the borrowers' revenues and the loan agreements are off-chain. The liabilities are not. Every deposit, loan disbursement, repayment, fee, redemption request and bridge transfer leaves an event log on a public chain. Those logs cannot prove that a GPU exists. They can show how much was lent, to how many borrowers, at what rates, how much has been repaid, whether the yield credited to holders is backed by cash, who holds the tokens and how quickly holders can exit. A depositor or a risk curator needs those numbers before relying on the off-chain story.

We rebuild USD.AI from raw logs using the Covalent GoldRush API [4] and the protocol's source code [5], and ask seven questions:

- RQ1 Where does deposited money go?
- RQ2 Where does the yield come from, and is NAV growth backed by cash from borrowers?
- RQ3 Who holds the yield-bearing token?
- RQ4 How concentrated are the borrowers?
- RQ5 How do redemptions behave, and do they follow GPU rental prices?
- RQ6 Which chains does the protocol live on?
- RQ7 How large is GPU credit compared with the compute actually bought and paid for on-chain, on Akash and Livepeer?

We also check every headline number against the vault's own accounting at a single block, and we trace one kind of vault outflow, escrow pre-funding, to where the money ended up.

Contributions.

1) An open, tested pipeline that decodes every event from source-code signatures, follows loans through a V1 to V2 migration and through refinancings, and reproduces all results offline from a hash-verified dataset.
2) A reconstruction of a GPU-backed loan book from chain data: 39 loans to 13 borrowers, $290.1 M originated and $182.2 M outstanding.
3) A cash-coverage test that separates yield received in cash from yield credited through NAV.
4) A reconstruction of the collateral from the protocol's hardware-token metadata: model, unit count, stated value and loan-to-value for every active loan.
5) Measurements of holder concentration, the redemption queue and supply across five chains.
6) A reconciliation showing that 23.2% of reported vault assets is an escrow claim on funds that were redeemed to PYUSD and sent onward before any matching loan existed.
7) A comparison of GPU credit with spending on two decentralized compute markets over the same period.
8) Monitoring indicators that lenders and curators can run with the same code.

Earlier empirical work on DeFi lending studied interest rates and liquidations in over-collateralized markets with on-chain collateral [6]–[8]. Stablecoin research models peg stability and run risk [9]–[11]. A GPU-backed dollar fits neither. Its collateral is off-chain and illiquid, its assets are term loans, and holders exit through a queue that an administrator services. Our earlier paper modeled the economic life of GPUs as assets [12]. This one looks at the debt financed against them.

II. Protocol Background

USD.AI issues two tokens on Arbitrum, its hub chain [13]. *USDai* is minted one-for-one against approved stablecoins (PYUSD, M^0's wM and USDC over our window) by KYC'd institutions through `deposit()` and burned back to a stablecoin through `withdraw()`. It does not accrue yield. *sUSDai* is a vault over USDai. Deposits are synchronous under ERC-4626 [14]. Exits are asynchronous under ERC-7540 [15]: a holder calls `requestRedeem()`, which burns the shares and joins a first-in, first-out queue. An administrator role then services the queue at fixed windows (the documentation gives 30 days as an example) by calling `serviceRedemptions()`. The vault prices deposits at an *optimistic* NAV and redemptions at a *conservative* NAV, and the two coincide only when no loans are active [13].

Vault capital reaches borrowers through a lending stack (Fig. 1). The vault commits USDai to a deposit or escrow timelock keyed by a `loanTermsHash`. A *Loan Router* originates the loan, mints lender position NFTs, and routes monthly repayments of principal and interest back to tranche lenders. Collateral is a wrapped bundle of GPU-backed tokens, and loans are non-recourse to the borrower's parent except for "bad acts" [13]. A second yield source, a *Base Yield Escrow*, holds PYUSD issuer incentives that are periodically harvested into the vault as newly minted USDai. Earlier in our window, yield on M^0's wM reserve was also minted into the vault as base yield. Both tokens are bridged with LayerZero burn-and-mint OFTs [16] to Ethereum, Base, Plasma and Solana. Every core contract can be paused by an administrator role, and operations are submitted by a multisig [13].

The source code and its full git history are public [5]. We use them in two places. First, every event signature we decode comes from that code, including signatures that were later renamed or removed. Second, the history shows a V1 to V2 Loan Router upgrade (commit `43dd6d5`, 2026-05-29) and a temporary `migrateLoan()` path, which we see executed on-chain between 2026-06-26 and 2026-07-14.

III. Data and Methodology

A. Data acquisition

All chain data comes from the Covalent GoldRush REST API [4] (Table I). The study window runs from the vault's deployment block (Arbitrum block 335,730,936, 2025-05-12) to block 517,316,432 (2026-10-22). The first vault deposit, and so the first NAV observation, is on 2025-05-14. Livepeer and Akash are measured over the same dates. We pull three kinds of data:

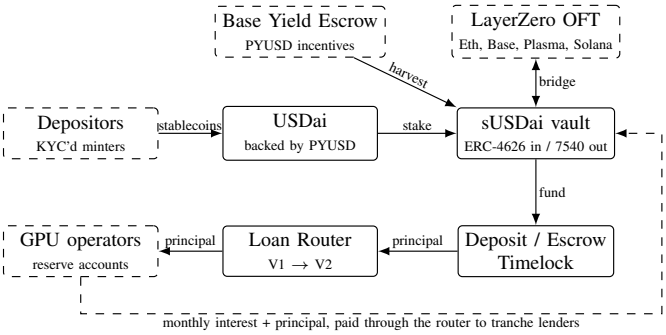


Fig. 1. USD.AI as reconstructed in this study. Solid arrows are flows we measure directly from Arbitrum event logs. Dashed boxes are parties or venues whose internal state is not on the hub chain.

- **Complete transaction histories**, with decoded logs, for the nine lending-stack contracts (Table I).
- **Event scans** of the USDai and sUSDai contracts for 15 event types. The topic endpoint accepts at most 10^6 blocks per call, so the window is split into 182 non-overlapping block windows. Windows that are fully in the past are cached; the open window is always refetched.
- **Snapshots**: current holders of USDai and sUSDai on four chains, and daily USD prices.

We also use the weekly median H100 on-demand rental price from a frozen GetDeploying snapshot (CC BY 4.0) [17], the same file used in [12]. Its SHA-256 matches the original.

B. Decoding

We decode raw logs ourselves rather than rely on third-party ABI coverage. For each event we compute $topic_0 = \mathrm{keccak256}(\textit{signature})$ from the published source. A log is accepted only if its topic count matches the number of indexed parameters, which, for example, rejects ERC-721 `Transfer` events. The data field is then ABI-decoded. V2 originations, migrations and refinancings carry the full `LoanTermsV2` struct: borrower, currency, collateral token IDs, tranche lenders, amounts and per-second rates, repayment schedule and fee specifications. We decode it to recover terms that V1 only kept in calldata. The decoder recognizes 55,954 events. Two topic hashes on the V1 router match no signature in the repository history; we count them and exclude them.

C. Stitching the loan book

A loan changes identity when it is migrated or refinanced, because each event retires one `loanTermsHash` and creates another. We build a forest in which each new hash points to its predecessor and resolve every repayment to the root origination. Migrated loans name a per-borrower *reserve account* as the new borrower. The factory event `ReserveAccountCreated(account, borrower)` maps each reserve account back to the underlying borrower. Each hash keeps the currency of its own terms: V1 loans were denominated in PYUSD or USDC (6 decimals) and migrated loans in USDai (18 decimals). Outstanding principal is the principal of the current terms

TABLE I
DATA SOURCES AND WHAT EACH ONE SUPPLIES

Source	Endpoint or method	What we pull
GoldRush	`transactions_v3`	Full decoded history of the 9 lending contracts: Loan Router V1 and V2, deposit, escrow and collateral timelocks, reserve-account factory, collateral wrapper, base yield escrow
GoldRush	`events/topics`	15 vault and USDai event types over 182 block windows; Livepeer `TicketBroker` ticket payments
GoldRush	`token_holders_v2`	Current holders of USDai and sUSDai on 4 chains; monthly sUSDai holders on Ethereum at past blocks
GoldRush	`historical_by_addresses_v2`	Daily market prices of sUSDai, USDai, PYUSD and WETH
GoldRush	`transactions_summary`	Activity of top holders and borrowers
GoldRush	`address/{a}/activity`	Every chain each borrower and top holder has used
GoldRush	`balances_v2`	Current holdings of traced wallets
Arbitrum node	`eth_call`	Vault accounting views at one block
Arbitrum node	`eth_getLogs`	Independent re-pull of every event set, to check GoldRush (Section IV-F)
Arbitrum node	`tokenURI`, `ownerOf`	Hardware-token metadata links and holders for every pledged token
USD.AI metadata host	HTTPS	The protocol's own record for each hardware token: model, units, stated value, useful life
Solana node	`getTokenSupply`	sUSDai and USDai supply on Solana
Akash Console API	REST	Akash spend, leased GPUs, providers, GPU prices
GetDeploying	frozen file	Weekly median H100 rental price (CC BY 4.0)

minus principal repaid under those terms. The tranche rate r is per-second and scaled by 10^{18}, so the annual rate is $\mathrm{APR} = r \cdot 31{,}536{,}000/10^{18}$. Loans below $1,000 are test loans and are excluded.

D. Vault accounting

Let $\mathcal{D}_d$ be the set of sUSDai `Deposit` events on day d. We define the deposit-side NAV and the global share supply as

$$\mathrm{NAV}_d = \mathrm{median}_{e \in \mathcal{D}_d} \frac{assets_e}{shares_e}, \tag{1}$$

$$S_d = \sum_{e \le d} shares_e^{\text{mint}} - \sum_{e \le d} shares_e^{\text{request}}. \tag{2}$$

NAV is forward-filled on days without deposits. Shares are burned when a redemption is requested, not when it is paid. Bridging moves shares between chains without changing S_d; we validate S_d against the sum of the four holder snapshots. The yield accrued to holders in a period P is

$$Y_P = \sum_{d \in P} S_{d-1} \left(\mathrm{NAV}_d - \mathrm{NAV}_{d-1}\right), \tag{3}$$

and the cash that reached the vault is $C_P = I_P + B_P + F_P + E_P$. Here I_P is loan interest paid to the vault (`LenderRepaid` with the vault as lender), B_P is base yield harvested into the vault, F_P is fees paid to the vault, and E_P is escrow interest credited on escrow withdrawal or cancellation, net of the 10% router admin fee. The *cash coverage ratio* $\kappa = C/Y$ asks how much of the NAV growth holders were credited with arrived as cash. A ratio below one means part of the reported yield is accrued, not received. That is normal for loans that pay monthly, but it should close over time.

E. Holders, queue and GPU prices

For each chain we report holder count, top-k shares, the Herfindahl-Hirschman index (HHI) and the Gini coefficient. Top holders are labeled using the contract metadata names GoldRush returns for logs they emit. Redemption requests are matched to `RedemptionProcessed` events by request ID to obtain wait times and pending shares. Finally, we compute Spearman's ρ between weekly requested shares and the H100 rental price, both its level and its week-over-week change. The sample is small, so this is descriptive and not causal.

F. Reconciliation with on-chain state

Event-derived numbers can be wrong in ways that are hard to see from the events alone. We therefore read the vault's own accounting views (`totalAssets`, `totalShares`, `bridgedSupply`, `depositTimelockBalance`, `loanRouterBalances`, the deposit and redemption share prices, and related views) with `eth_call` at a single Arbitrum block, 517,316,432. GoldRush does not expose arbitrary contract calls, so this one step uses the public Arbitrum RPC. Every reconciliation in Section IV-E compares an independently computed event total to one of these views.

G. Following escrowed funds

The vault can commit USDai to an escrow timelock against a `loanTermsHash` before any loan with that hash exists.

When an escrow deposit has no matching origination, we follow the stablecoins. Starting from the escrow facility that received the USDai, we pull complete transaction histories with GoldRush and decode every ERC-20 `Transfer` of USDai, PYUSD and USDC, plus USDai `Withdrawn` events, which mark redemption of USDai into PYUSD. We expand two hops and stop at wallets with more than 1,500 transactions, which we treat as services. We label wallets only by their on-chain role. We do not attribute them to off-chain entities.

H. Decentralized compute markets

To put GPU credit next to realized on-chain demand for compute, we measure two live markets. *Livepeer* pays for video transcoding and AI inference with probabilistic micropayments on Arbitrum. A gateway funds a deposit in the `TicketBroker` contract and an orchestrator redeems winning tickets. Each `WinningTicketTransfer(sender, recipient, amount)` event is ETH that actually moved from a payer to a provider. We scan it over the same window and price it at the daily WETH price from GoldRush. *Akash* runs on its own Cosmos chain, which GoldRush does not index. We take its daily USD spend, active GPU count, provider count and per-model GPU prices from the public Akash Console API, and label them as such. Finally, we compare GPU rental prices on Akash with the centralized-cloud H100 median in the GetDeploying snapshot.

I. Where each number comes from

Event logs, transaction histories, holder lists, prices, cross-chain activity and wallet balances come from GoldRush. The vault's accounting views at one block come from `eth_call` on a public Arbitrum node, Solana token supply from a public Solana node, Akash figures from the Akash Console API, and GPU rental prices from GetDeploying. Section IV-F re-checks the GoldRush event data against a public node.

J. Reproducibility

Raw responses are canonicalized into `data/frozen` with a SHA-256 manifest. `reproduce.py` regenerates every number, table and figure in this paper offline; numbers enter the text through generated LaTeX macros. The repository has 40 unit tests, including tests that the API key cannot leak, and runs them in CI.

IV. Results

Fig. 2 lists the main measured quantities. We take the research questions in order.

A. RQ1: Where the money goes

Over the window, institutions minted \$1,653.8 M of USDai and burned \$1,400.2 M. Minting was backed by PYUSD, by M^0's wrapped \$M (wM), by USDC and, marginally, by USDT (Table II). Not every burn is an exit. Loan disbursements also burn USDai into the borrower's currency at origination, and the escrow redemption in Section IV-G is a burn too. Of the USDai still outstanding, most sits in the vault as idle deposit balance rather than in loans (Section IV-E).

TABLE II
USDai minting and redemption by backing asset

Backing asset	Minted	Redeemed
PYUSD	\$757.80 M	\$1,140.29 M
wM (M^0)	\$542.73 M	\$33.95 M
USDC	\$353.25 M	\$225.99 M
USDT	\$0.04 M	\$0.00 M
Total	\$1,653.8 M	\$1,400.2 M

From that pool the Loan Routers originated 39 loans (excluding 2 test loans below \$1,000) for \$290.1 M of principal to 13 distinct borrowers. 17 loans were moved from V1 to V2 through `migrateLoan()` and are counted once. 18 loans have been repaid in full. None has a breach or liquidation event. At 2026-10-22, \$182.2 M was outstanding. Borrowers have paid \$6.94 M of interest, \$108.1 M of principal and \$515 k of repayment fees. Fig. 3 shows originations by month and borrower.

B. What the collateral is

Every loan's collateral is an ERC-721 token from the protocol's hardware contract ("USD.AI Tokenized Hardware"), either directly or wrapped in a bundle. Each token's `tokenURI`, read from the contract, points to a metadata record the protocol publishes with the server model, the number of units, a stated collateral value, a useful life and a depreciation start. We read the token IDs from every decoded `LoanTermsV2` and from the bundle wrapper's transfer logs, the URIs and owners by `eth_call`, and the records from the protocol's host, 87 tokens in all (retrieved 2026-10-27). The records are the protocol's own statements, hosted off-chain and changeable, so what follows is what USD.AI says the collateral is, not an appraisal.

Table III and Fig. 4 list the active loans. The hardware is current generation: NVIDIA B300 and B200 servers, H200 servers and, in the small early loans, RTX 6000 and RTX 5090 workstations. Counting eight GPUs per B200, B300 and H200 server (an assumption the one press-reported loan confirms exactly, Section IV-I), the active book is secured by 630 units carrying about 4,560 GPUs with a stated value of \$272.8 M, against \$182.2 M outstanding: a book loan-to-value of 66.8%, ranging from 43% to 80% by loan. Every record uses a useful life of 1080 and 1095 days, three years, with depreciation starting "on borrow". That is the assumption our earlier work questioned [12]: at a three-year straight line, a loan at 80% LTV is under-collateralized within eight months unless the stated value already discounts the hardware.

One token stands out. NVIDIA B300 (288), 288 servers with a stated value of \$140.2 M, secured the largest borrower's \$87.1 M loan until that loan was repaid in full on 2026-10-01, eight days after the \$128.9 M escrow deposits. The token now sits unpledged in the borrower's reserve account, and a further record for 32 GB200 NVL72 racks (\$153 M stated) exists on the metadata host without a minted token. Read together with Section IV-G, the simplest explanation is a refinancing

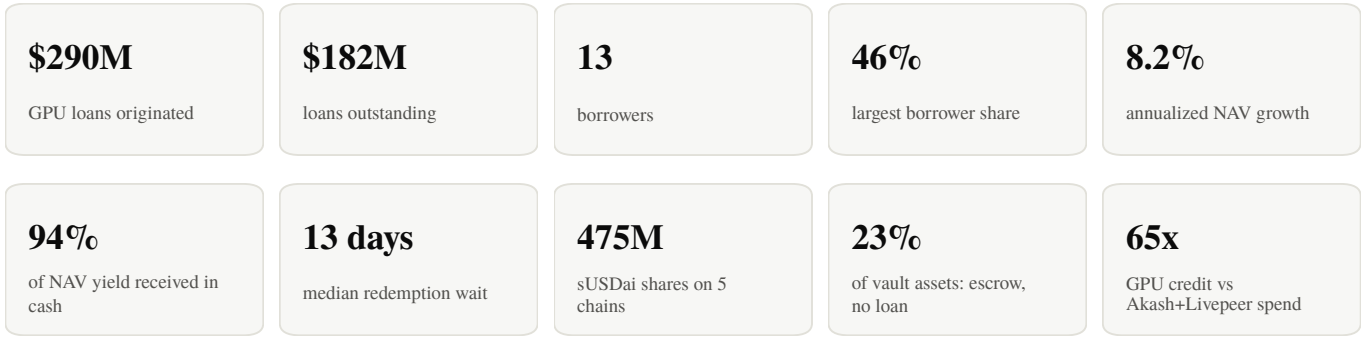


Fig. 2. Key quantities at a glance, all computed from chain data over 2025-05-12 to 2026-10-22.

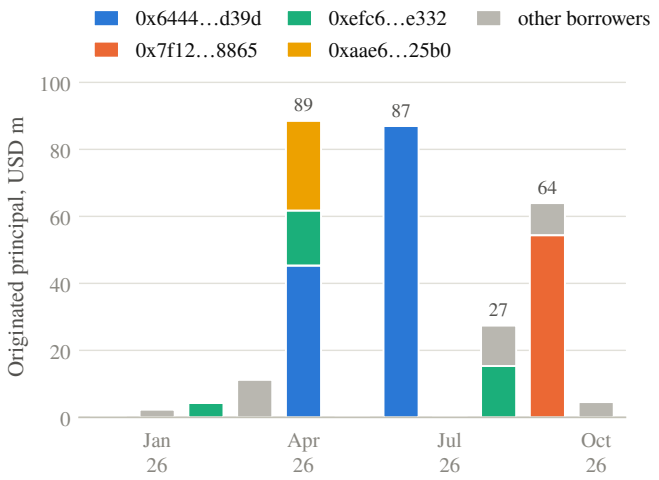


Fig. 3. Monthly originated principal by borrower. The four largest borrowers are shown separately.

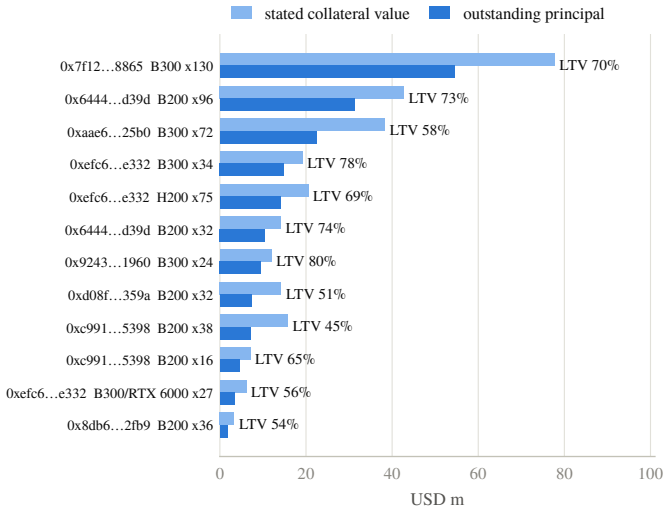


Fig. 4. Stated collateral value against outstanding principal for the largest active loans.

TABLE III
COLLATERAL BEHIND THE ACTIVE LOANS, FROM THE PROTOCOL'S HARDWARE-TOKEN METADATA. GPUS COUNTED AT EIGHT PER B200, B300 OR H200 SERVER; WORKSTATION UNITS NOT COUNTED.

Borrower	Hardware (units)	GPUs	Stated value	Outstanding	LTV
0x7f12...8865	B300 (130)	1,040	$77.7 M	$54.4 M	70%
0x6444...d39d	B200 (96)	768	$42.7 M	$31.2 M	73%
0xaae6...25b0	B300 (72)	576	$38.3 M	$22.4 M	58%
0xefc6...e332	B300 (34)	272	$19.2 M	$14.9 M	78%
0xefc6...e332	H200 (75)	600	$20.5 M	$14.1 M	69%
0x6444...d39d	B200 (32)	256	$14.1 M	$10.5 M	74%
0x9243...1960	B300 (24)	192	$12.0 M	$9.6 M	80%
0xd08f...359a	B200 (32)	256	$14.2 M	$7.3 M	51%
0xc991...5398	B200 (38)	304	$15.8 M	$7.1 M	45%
0xc991...5398	B200 (16)	128	$7.1 M	$4.6 M	65%
0xefc6...e332	B300/RTX 6000 (27)	72	$6.1 M	$3.4 M	56%
0x8db6...2fb9	B200 (36)	64	$3.3 M	$1.8 M	54%
0x6726...be31	H200 (8)	32	$0.9 M	$0.4 M	43%
8 smaller loans	mixed (10)	n/a	$0.97 M	$0.59 M	60%
All active	(630)	4,560	$272.8 M	$182.2 M	67%

in progress: the old loan repaid, the new one funded through escrow but not yet originated. That does not change what holders are exposed to while it is pending, but it is the reading the protocol is most likely to give.

C. Maturities and payment status

The 21 active loans have original terms of 2.1 to 3.0 years. Weighted by outstanding principal, the book has 2.6 years left to run: $13.3 M matures in 2028 and $168.9 M in 2029. Against that, holders exit through a queue with a 13-day median wait (Section IV-J). Every active loan is current. The longest gap since a loan's last repayment is 25 days, inside the monthly schedule, and 0 loans are late. The oldest active loan is under a year old, so the book has not yet been through the part of a GPU loan's life where hardware has lost most of its value.

D. RQ2: Where the yield comes from

The deposit-side NAV of sUSDai rose from 1.0003 to 1.1202 (Fig. 5). That is 8.17% annualized over the full window and 7.60% over the last 90 days. Multiplying daily NAV changes by the outstanding share supply gives $24.46 M of yield credited to holders.

Against that, the vault received $6.65 M of loan interest, $15.08 M of base yield (net of a $1.68 M admin fee), $305 k of origination and repayment fees, and $880 k of escrow interest

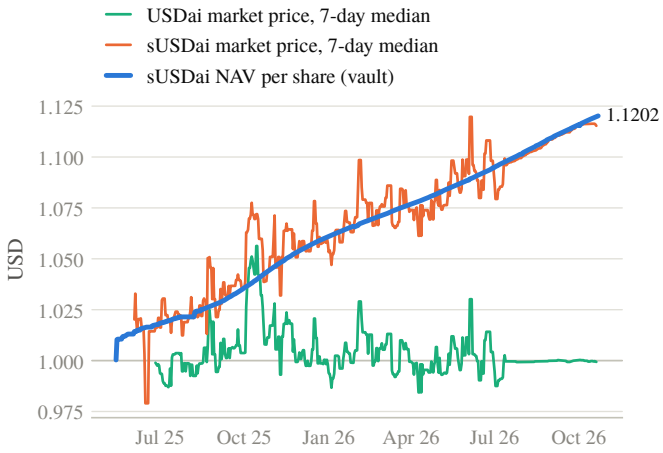


Fig. 5. sUSDai NAV per share from vault deposits and the USDai market price (GoldRush pricing).

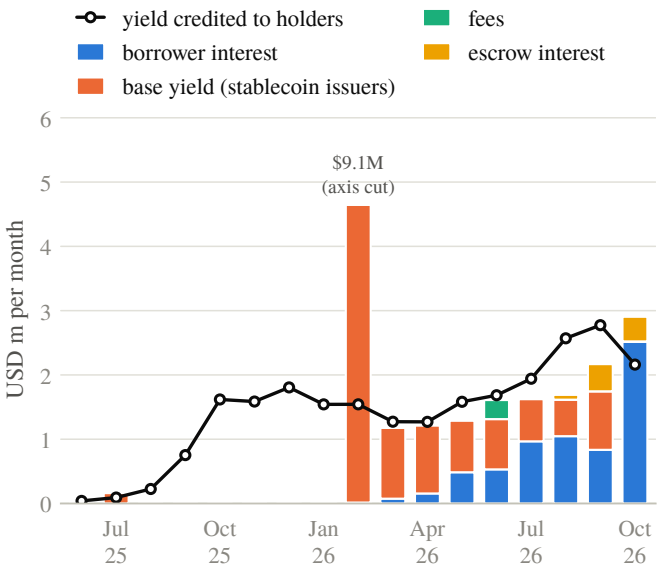


Fig. 6. Yield credited to holders (NAV growth × supply) against cash received by the vault, by month.

(net of the router admin fee). Cash coverage is $\kappa = 93.7\%$. Counting only what borrowers paid (interest and fees), it is 28.4%. The gap is consistent with interest that accrues daily but is paid monthly, plus base yield accrued but not yet harvested. Fig. 6 shows the monthly split. Base yield was the largest cash source, at $15.08 M against $6.65 M of borrower interest. Base yield has two sources, which we separate by the other events in each transaction. $6.36 M came from PYUSD issuer incentives harvested out of the Base Yield Escrow (a `USDai.Harvested` event in the same transaction). $8.72 M was minted from M^0's wM, the yield-bearing stablecoin that backed USDai early on, including a single $7.82 M deposit on 2026-02-02. Each source on its own is smaller than borrower interest. Combined, they are more than twice as large, so over this window sUSDai holders received more cash from stablecoin issuers than from GPU borrowers.

Interest also goes to lenders other than the vault. Of the interest paid, $6.65 M (95.8%) went to sUSDai. $294 k went to two other tranche lenders: a junior tranche priced at 28% APR and a one-unit senior position that collects repayment fees. Loans are almost entirely vault-funded (99.95% of current principal).

E. Reconciliation with vault state

At block 517,316,432 the vault reports $555.4 M of assets against 495.7 M shares (Fig. 7). Shares split into 270.8 M on Arbitrum, 204.8 M bridged out and a residual of 20.17 M, which the contract counts as pending redemption. Our own redemption queue shows 20.17 M pending, so the two differ by under 1 M shares. Event-derived supply plus our pending queue matches `totalShares` to within 0.00%; part of that closeness comes from offsetting small differences. The bridged supply matches the sum of the three EVM spoke snapshots plus Solana (204.6 M). The vault's `pendingLoanBalance` is $181.8 M. Our stitched loan book differs from it by $0.39 M (0.22%). We take these matches as validation of the decoder, the stitching and the unit handling.

The assets side breaks down as $242.3 M of idle deposit balance, $181.8 M of active loans, $1.04 M of accrued loan interest, $0.98 M of unharvested base yield, $0.66 M of accrued escrow interest and $128.9 M of `depositTimelockBalance`. That last item is 23.2% of assets, and it is not a loan.

F. Checking GoldRush against the chain

Almost every number in this paper comes from GoldRush, so we pulled the underlying logs again from a public Arbitrum node with `eth_getLogs`, without GoldRush, over the same block range. For each of the 22 event sets the analysis depends on, we compare the two sources log by log, identifying each log by its transaction hash, topics and data, and we compare the sum of the key amount (Table IV). 22 of 22 sets are identical, covering 246,262 logs.

The check found one real problem. GoldRush's event endpoint treats `ending-block` as exclusive, so our first scan skipped the last block of every 1,000,000-block window. We confirmed this directly: a log in block B is returned for the range $[B, B+1]$ but not for $[B, B]$. We fixed the request, fetched the 182 window-end blocks for every scanned event and re-ran the check. The patch added two Livepeer logs (one ticket payment of 0.0024 ETH) and nothing to the USD.AI events. We also found that, for some transactions, GoldRush numbers logs differently from the node, so we never match or deduplicate logs on their index.

For token supply, `totalSupply` at the snapshot block equals the sum of GoldRush holder balances exactly on Base and is 0.19% higher on Plasma. The public Arbitrum and Ethereum nodes do not serve historical state, so there we compare against the latest block, a few hours after the snapshot, and the gaps are below 0.1%. The escrow redemption receipt,

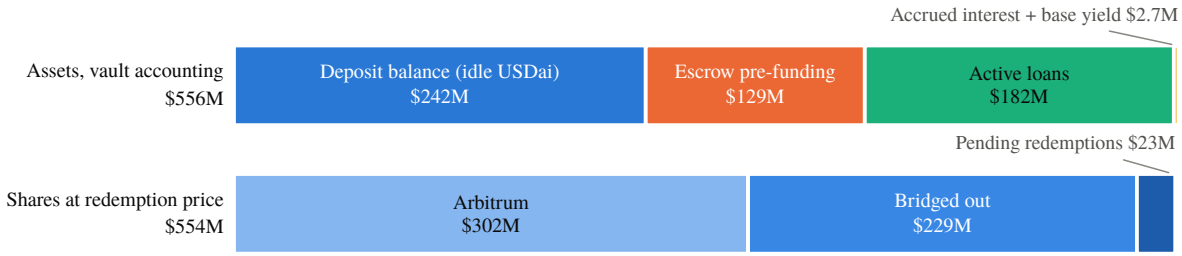


Fig. 7. The vault's balance sheet at block 517,316,432, read with `eth_call`. Shares are valued at the redemption share price.

TABLE IV
GOLDRUSH DATA CHECKED AGAINST A PUBLIC ARBITRUM NODE

Event (Arbitrum)	RPC logs	GoldRush logs	Same
Loans originated, router V1	35	35	yes
Loans originated, router V2	6	6	yes
Loans migrated V1 to V2	17	17	yes
Loans refinanced	3	3	yes
Loans refinanced (3-field)	2	2	yes
Repayments, router V1	118	118	yes
Repayments, router V2	79	79	yes
Lender payouts, router V1	235	235	yes
Lender payouts, router V2	96	96	yes
Fees paid, router V2	61	61	yes
Breaches and liquidations, V2	0	0	yes
Reserve accounts created	25	25	yes
USDai mints	16,377	16,377	yes
USDai redemptions	8,867	8,867	yes
sUSDai deposits	20,093	20,093	yes
sUSDai redemption requests	3,002	3,002	yes
sUSDai redemptions processed	2,841	2,841	yes
Base yield deposited to vault	12	12	yes
Escrow deposits	9	9	yes
Escrow withdrawals	4	4	yes
Escrow cancellations	3	3	yes
Livepeer winning-ticket payments	194,377	194,377	yes
sUSDai supply: `totalSupply` vs. sum of GoldRush holders			
Arbitrum (latest block)	268.98 M	270.82 M	+0.68%
Ethereum (latest block)	102.14 M	101.53 M	-0.60%
Base (same block)	3.64 M	3.64 M	+0.00%
Plasma (same block)	26.16 M	26.11 M	-0.19%

re-read from the node, shows the same burn of 128,895,244.1 USDai and the same PYUSD payment to the same wallet. Finally, USDai itself is backed: at block 518,591,558 the USDai contract held $266.3 M of stablecoins ($266.3 M of it PYUSD) against $266.3 M of USDai across the hub and the EVM spokes, a coverage of 100.00%. The idle deposit balance in the vault is therefore real PYUSD one redemption away, not a claim on a claim.

G. Escrow pre-funding

The vault made 9 escrow timelock deposits totaling $223.1 M. Most closed normally: 7 were either withdrawn into an originated loan or cancelled (3 cancellations, which returned principal plus interest to the vault). 2, both made on 2026-09-23 and worth $128.9 M, carry a `loanTermsHash` that no Loan Router has originated and that has not been cancelled. Together they equal `depositTimelockBalance` to the cent. They earn an escrow rate of 6.5%, which the vault accrues optimistically. None of the timelock contracts holds any tokens: `balanceOf` returns zero for USDai, PYUSD and USDC.

Fig. 8 and Table V show where the tokens went. Every escrow deposit moves USDai from the vault to a single address we call the escrow facility ($223.1 M in total, of which cancellations returned part). It is a Gnosis Safe, and its transactions are submitted by the same accounts that submit the protocol's own multisig transactions, so it appears to be operated by the protocol side, not by a borrower. For the two open deposits, the facility redeemed the USDai into PYUSD at the USDai contract the same day and paid the PYUSD to a single wallet we call the PYUSD hub. The hub was first active on 2026-04-29 and has 56 transactions. It also received PYUSD from the largest borrower and from the account that submits the protocol's multisig transactions. It forwarded almost everything to a high-activity wallet (off-ramp A) that holds PYUSD and USDG and is active on 19 chains, and a smaller amount to a service wallet. The fee recipient that collects origination fees routes to the same off-ramp.

Escrow pre-funding is allowed by the contracts, and it may correspond to a signed facility whose loan has not yet been put on-chain. But from the chain's point of view, $128.9 M of the assets backing sUSDai is a claim that some party will later post a matching loan. The funds behind it have left the protocol's contracts, passed through a Safe run by the same operators, and moved to an off-ramp wallet. Holders are paid optimistic yield on that claim before any GPU collateral is wrapped. We describe wallets only by their on-chain roles, and the protocol should have the chance to explain the arrangement (Section V).

H. RQ3 and RQ6: Holders and the cross-chain footprint

On Arbitrum, sUSDai has 3,778 holders. The largest holds 33.3% and the top ten hold 86.7% (HHI 0.209, Gini 0.994). Table VI gives the same statistics for each chain, and Table VII lists the largest hub holders with the contract labels GoldRush returns. On every chain, a handful of contracts and pools hold most of the supply (Fig. 9). The largest DeFi venue is Pendle, which holds 13.2% of Arbitrum sUSDai. On Plasma, 44.1% of USDai sits in Pendle and another 20.0% in vault strategies,

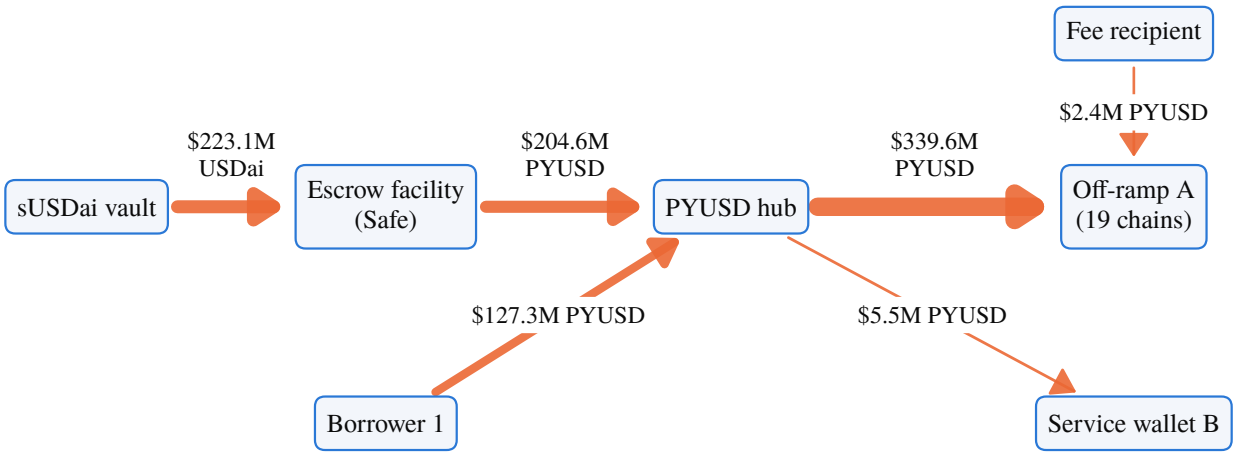


Fig. 8. Where escrowed vault funds went. Edge width is proportional to USD. All flows are decoded ERC-20 transfers or USDai redemptions on Arbitrum. Wallets are labeled by on-chain role only.

TABLE V
LARGEST STABLECOIN FLOWS AROUND THE ESCROW FACILITY AND THE PYUSD HUB. HUB OUTFLOWS ALSO INCLUDE PYUSD THE HUB RECEIVED FROM THE LARGEST BORROWER AND THE MULTISIG SUBMITTER.

From → To	Asset	USD
sUSDai vault (0x0b2b...5ef9) → escrow facility	USDai	$223.1 M
PYUSD hub (0x1168...21b1) → off-ramp A (0xf845...abb7)	PYUSD	$339.6 M
escrow facility (0x844b...6286) → PYUSD hub (0x1168...21b1)	PYUSD (redeemed)	$142.4 M
escrow facility (0x844b...6286) → PYUSD hub (0x1168...21b1)	PYUSD	$62.2 M
escrow facility (0x844b...6286) → sUSDai vault (0x0b2b...5ef9)	USDai	$16.3 M
escrow facility (0x844b...6286) → 0x24a9...1392	PYUSD	$6.0 M
escrow facility (0x844b...6286) → 0xfd1c...ca91	USDai	$5.9 M
0xfd1c...ca91 → PYUSD hub (0x1168...21b1)	PYUSD (redeemed)	$5.9 M
PYUSD hub (0x1168...21b1) → service wallet B (0x264b...97b5)	PYUSD	$5.5 M
escrow facility (0x844b...6286) → 0x6034...23ef	USDai	$5.1 M

so most of that chain's supply is in yield-trading and looping positions rather than in wallets.

Supply is spread across five chains: 270.8 M on Arbitrum, 101.5 M on Ethereum, 26.1 M on Plasma, 3.64 M on Base and 73.3 M on Solana (Fig. 10). Only one address holds at least 100,000 sUSDai on more than one EVM chain. It is a high-activity contract with more than 8 million Arbitrum transactions, and it is the largest or second-largest holder on all four EVM chains, with >95% of supply on Base and Plasma. Historical snapshots at monthly blocks show when the Ethereum footprint appeared. sUSDai on Ethereum was a test-sized position (2 holders in 2025-07-01, under 0.1 M shares on 2026-06-01) until mid-2026. It then grew to 89.3 M shares and 102 holders by 2026-10-01 (Fig. 11). Two addresses hold 99.7% of it: the cross-chain contract above and the Morpho Blue singleton (18.0%), where sUSDai is posted as loan collateral. GoldRush's cross-chain activity endpoint shows where borrowers and top holders operate (Fig. 12). The 10 borrowers with activity data have used between 4 and 12 chains (median 7), so borrowers are active well beyond Arbitrum. The 66 top holders range from 1 to 29 chains (median 10). Many single-chain holders are pools and vault contracts.

TABLE VI
SUSDAI HOLDER CONCENTRATION BY CHAIN

Chain	Holders	Supply	Top-1	Top-10	HHI
Arbitrum	3,778	270.8 M	33.3%	86.7%	0.209
Base	468	3.6 M	95.1%	99.6%	0.905
Ethereum	135	101.5 M	81.6%	100.0%	0.699
Plasma	1,424	26.1 M	97.6%	99.9%	0.953

TABLE VII
LARGEST SUSDAI HOLDERS ON ARBITRUM

Holder	sUSDai	Share	Label
0x7ced...20af	90.16 M	33.3%	unlabeled
0x52aa...e497	75.43 M	27.9%	unlabeled
0x30cc...2f62	35.68 M	13.2%	SY Staked USDai
0xed41...6029	8.89 M	3.3%	unlabeled
0xe7f5...e8e2	6.60 M	2.4%	unlabeled
0x9167...6700	5.38 M	2.0%	unlabeled
0xbeed...6dba	4.56 M	1.7%	unlabeled
0xa520...a575	2.82 M	1.0%	unlabeled

I. RQ4: Borrower concentration

The loan book is concentrated. The largest borrower accounts for 45.6% of originations and the top three for 76.8%. HHI is 0.273 on originations and 0.197 on outstanding principal. Both are above 0.18, the threshold for a highly concentrated market in the 2023 U.S. Merger Guidelines. Table VIII lists each borrower. Rates run from 7.0% to 15.0%, with an outstanding-weighted average of 10.26%. 25 reserve accounts map back to borrowers through the factory. Press coverage names some borrowers. One reported $26.82 M facility, secured by 576 NVIDIA B300 GPUs [18], matches, to the reported precision, a single loan of $26,821,644 originated on 2026-04-06, and its collateral record lists 72 B300 servers, which at eight GPUs per server is the same 576. We do not attribute any address to an off-chain entity in our results.

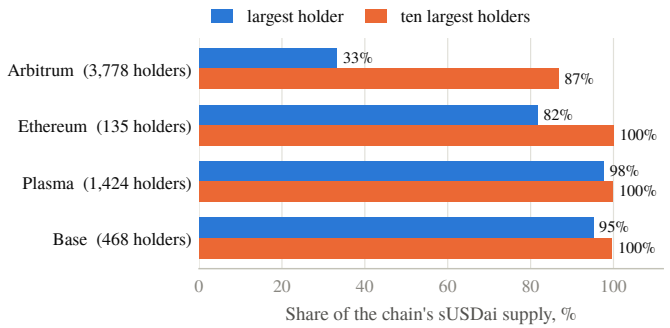


Fig. 9. Share of each chain's sUSDai held by its largest holder and its ten largest holders.

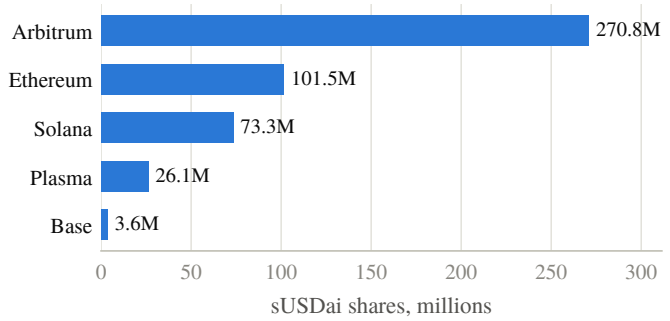


Fig. 10. sUSDai shares by chain (holder snapshots; Solana from mint supply).

J. RQ5: Redemptions and GPU prices

Holders filed 3,002 redemption requests from 2,545 addresses, for 350.3 M shares. The median completed request waited 13.0 days, the 90th percentile 26.6 days, and the longest 127.3 days. 20.17 M shares are still in the queue. Servicing is lumpy (Fig. 13): the queue is cleared in a small number of large batches. Three servicing calls on 2026-03-07 and 2026-04-06 together paid out about $217 M. Priced in PYUSD, which removes feed noise common to both tokens, USDai never traded below 0.975 (on 2025-08-19). Selling on a DEX is the only alternative to waiting.

We find no useful relationship between requests and GPU rental prices. Over 53 overlapping weeks, Spearman's ρ between weekly requested shares and the H100 rental price is -0.41 in levels and 0.01 in weekly changes (Fig. 14). The level correlation reflects two trends: redemptions grew as the protocol grew, while rental prices fell. It is not evidence that holders react to GPU prices.

K. RQ7: GPU credit versus realized compute demand

Over the same window, Livepeer gateways paid orchestrators 371 ETH ($1.01 M) in 194,377 winning tickets. 21 gateways paid 104 orchestrators. The largest gateway accounts for 49.4% of payments and the top three for 83.1% (HHI 0.318). On the supply side, payments are spread out (HHI 0.050). Livepeer's payments mix video transcoding and AI inference, and the events do not separate the two.

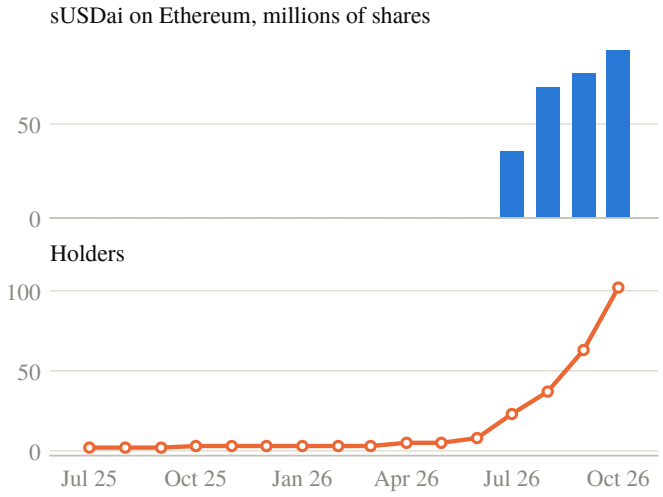


Fig. 11. sUSDai on Ethereum over time, from historical holder snapshots at the first block of each month.

TABLE VIII
BORROWERS (ADDRESS PREFIXES), SORTED BY ORIGINATED PRINCIPAL

Borrower	Loans	Originated	Outstanding	Interest paid
0x6444...d39d	3	$132.40 M	$41.71 M	$2,995 k
0x7f12...8865	1	$54.40 M	$54.40 M	$356 k
0xefc6...e332	3	$36.01 M	$32.42 M	$1,373 k
0xaae6...25b0	1	$26.82 M	$22.35 M	$1,123 k
0xc991...5398	3	$18.93 M	$11.70 M	$585 k
0x9243...1960	1	$9.56 M	$9.56 M	$58 k
0xd08f...359a	1	$7.50 M	$7.32 M	$121 k
0x8db6...2fb9	1	$2.30 M	$1.79 M	$206 k
0xe277...0eb9	7	$0.54 M	$0.43 M	$39 k
0x6726...be31	1	$0.53 M	$0.40 M	$31 k
0x7c5e...854e	1	$0.48 M	$0.00 M	$23 k
0x7b79...4462	15	$0.44 M	$0.00 M	$15 k
0x45b3...94d3	1	$0.20 M	$0.16 M	$15 k

Akash, measured through its own API, recorded $3.47 M of spend in the window and $6.5 M all-time. Active leased GPUs ranged from 40 to 529. On 2026-10-22, the network reported 263 of 390 GPUs active across 58 providers (Fig. 17). The median Akash H100 price was $2.56/hr and the A100 $1.49/hr, against a centralized-cloud H100 median of $3.36/hr (Fig. 18).

Fig. 16 and Table IX put the three side by side on a log scale. GPU credit lent through one on-chain vault is about 65 times all compute bought and settled on these two established decentralized compute markets combined ($4.47 M). Demand is concentrated on both sides of the comparison: USD.AI borrowers have an HHI of 0.273 and Livepeer gateways 0.318, while Livepeer's supply side is far more spread out (Fig. 19). None of the GPU use that these loans finance appears on-chain.

L. What the debt implies about the compute

The chain does not show the GPUs, but it shows what they have to earn. Each active loan has an outstanding principal, a fixed rate and a maturity, so the annual debt service is known: $18.7 M of interest plus $71.2 M of principal if the loans amortize straight-line to maturity, $89.9 M in all. Our

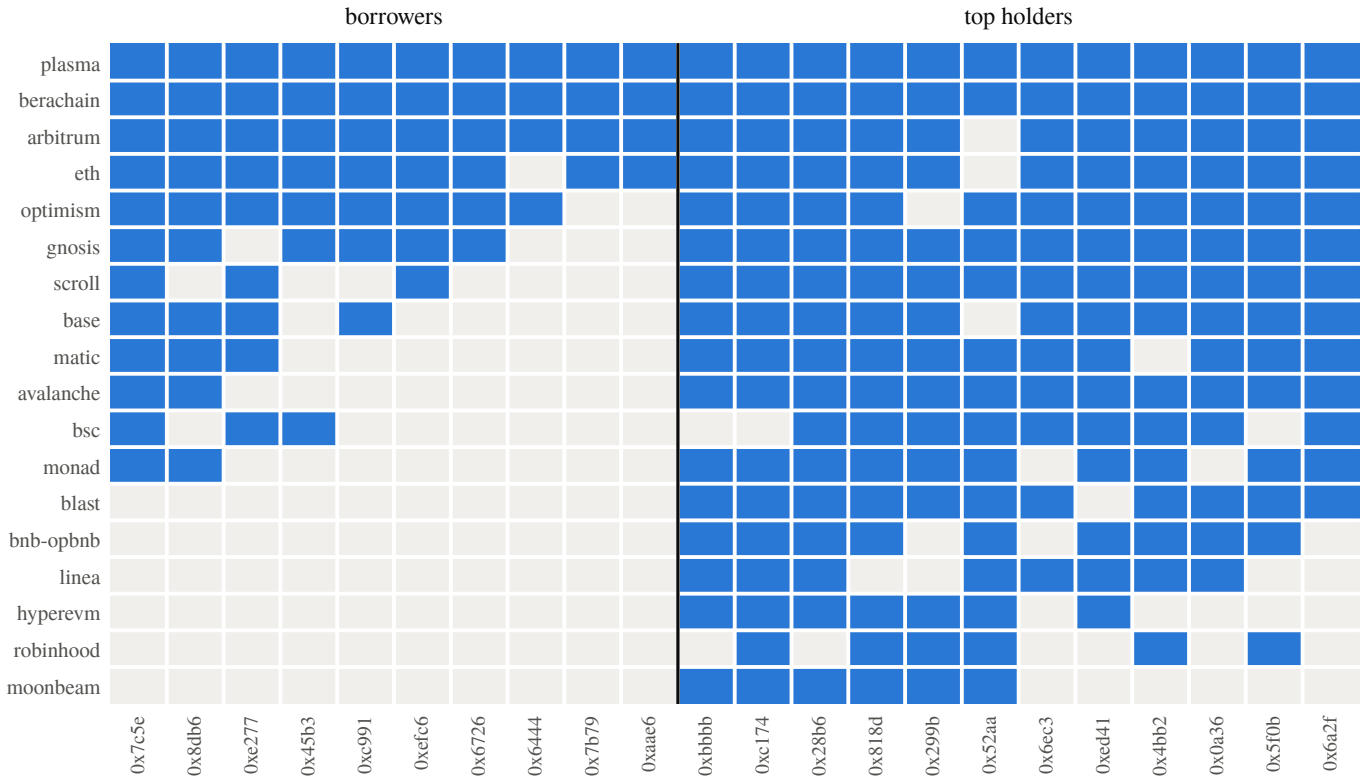


Fig. 12. Cross-chain activity of borrowers and top holders from GoldRush's address activity endpoint. Each filled cell is a chain on which the address has transacted.

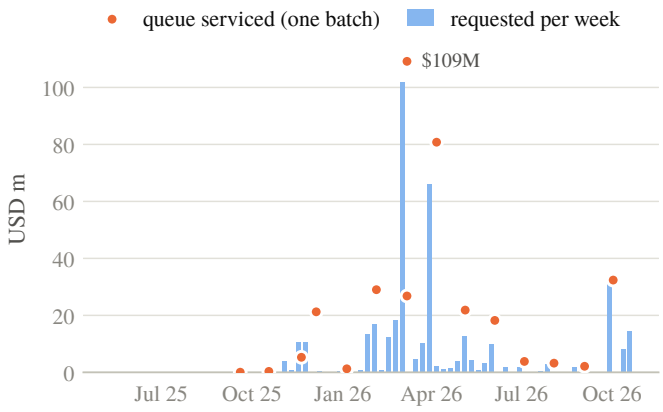


Fig. 13. Weekly redemption requests (USD at NAV) and queue servicing batches.

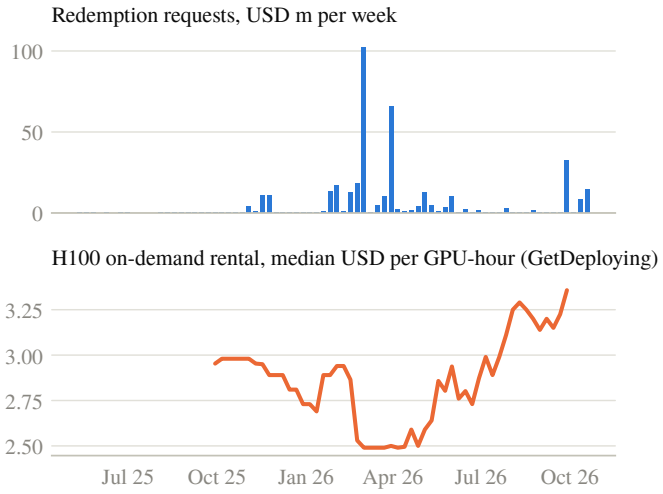


Fig. 14. Weekly redemption requests against the median H100 on-demand rental price (GetDeploying).

earlier work treated machine labor as hours of work rather than units of hardware [12], [19]; the same frame turns this debt into GPU-hours. At the Akash H100 median of $2.56 per hour, the borrowers must sell 35.1 M GPU-hours a year to service the debt from rental revenue alone. At 70% utilization that is 5,729 H100s running all year, drawing about 8.6 MW at 1.5 kW per GPU before cooling. At the centralized-cloud price of $3.36 it is 4,365 GPUs (6.5 MW); at $1.50, the low end of 2025 spot pricing [20], it is 9,778 (14.7 MW). Fig. 15 shows the curve. For scale, the entire Akash network has 390 GPUs. This is a lower bound: it ignores the borrowers' power, staff and data-center costs, and it assumes rental income is the only source of repayment. The pledged fleet is about 4,560 GPUs (Section IV-B), so at 70% utilization it covers debt service at $3.22 per GPU-hour and at full utilization at $2.25. Those are B200 and B300 class parts, which rent above H100 rates

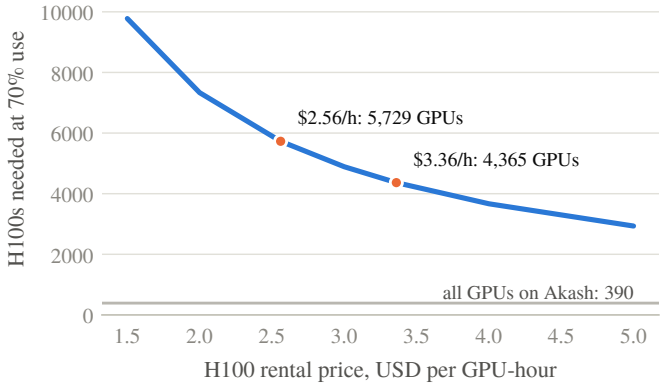


Fig. 15. H100s that must run at 70% utilization to service the outstanding debt from rental revenue, as a function of the rental price. The Akash network's total GPU count is shown for scale.

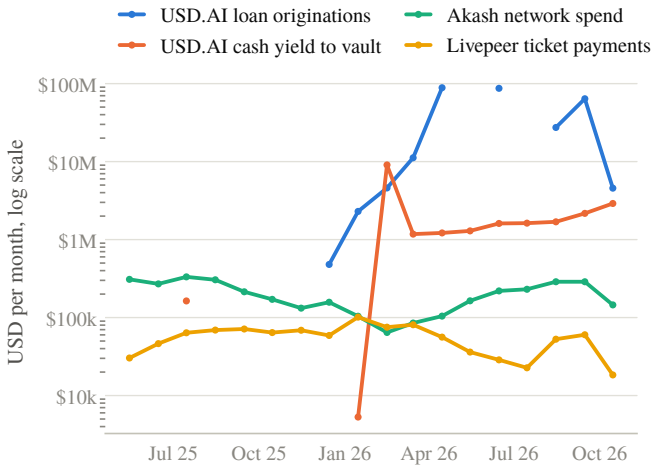


Fig. 16. Monthly USD volume: USD.AI originations and cash yield, Akash spend and Livepeer ticket payments (log scale).

today, so the fleet can carry the debt if it is rented; the margin is thinner than it looks, because the price has to cover power and operations as well, and the stated collateral value already assumes a three-year life.

The number is testable because the pledge is published. What is not published is utilization, rental revenue or the location and power contract of the hardware. A borrower that posted those would be the first a lender could underwrite from the chain alone.

V. Discussion

A. What the chain can and cannot tell a lender

Chain data is strong on liabilities and cash. It gives an exact loan book, every borrower address, every payment, every holder and every queued exit, and it matches the vault's own accounting to within a fraction of a percent. It is weak on assets. It cannot show that a GPU exists, that it is rented, at what price, or that a borrower earns enough to pay. Three of our findings sit at that boundary. First, more of the cash yield

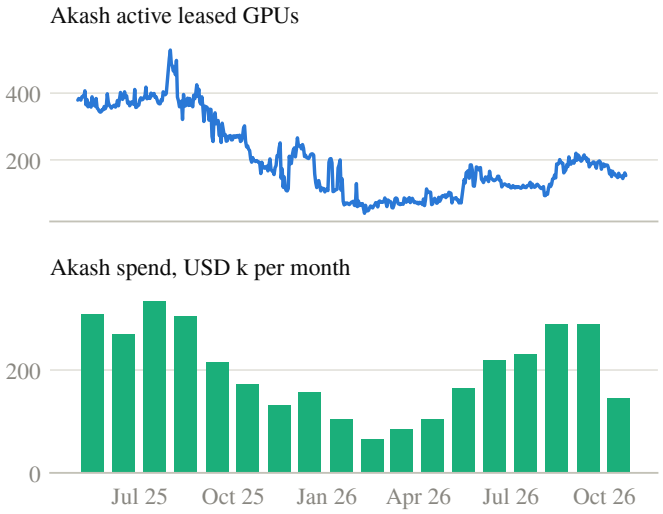


Fig. 17. Akash active leased GPUs and monthly spend (Akash Console API).

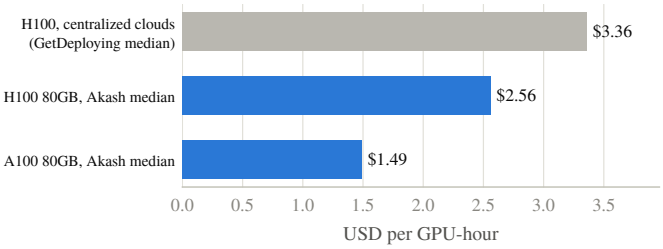


Fig. 18. GPU rental prices: decentralized (Akash median) and centralized (GetDeploying median), USD per GPU-hour.

has come from stablecoin issuers (M^0 and PYUSD) than from borrowers, so holders' returns depend on those programs as much as on GPU economics. Second, 23.2% of assets is an escrow claim on funds that left the protocol's contracts before any collateral was posted. Third, the loans run 2.6 more years on average while holders can queue to exit in two weeks, so the vault relies on new deposits and idle cash to pay exits. The contracts permit all three, and none shows up in the headline NAV.

B. What a neocloud unwind would do

The borrowers are GPU operators, the class of company now called neoclouds. That sector carries more than $20 B of GPU-backed debt from lenders such as Blackstone, PIMCO, Carlyle and Macquarie [20], and its stress is visible: CoreWeave's five-year credit default swaps traded at distressed levels in October 2026 [21]. H100 rental prices fell from $7 to $10 per hour in early 2024 to $2 to $4 by late 2025 [20], and the GetDeploying series in this paper puts the centralized median at $3.36. The debt structures assume GPUs keep enough value over five to six years to support the loans, while some operators and investors argue the economic life is two to three years [20].

USD.AI is a small on-chain slice of the same trade, with the same collateral and the same borrowers' customers. Table X shows what defaults would do to the vault, using the reconciled

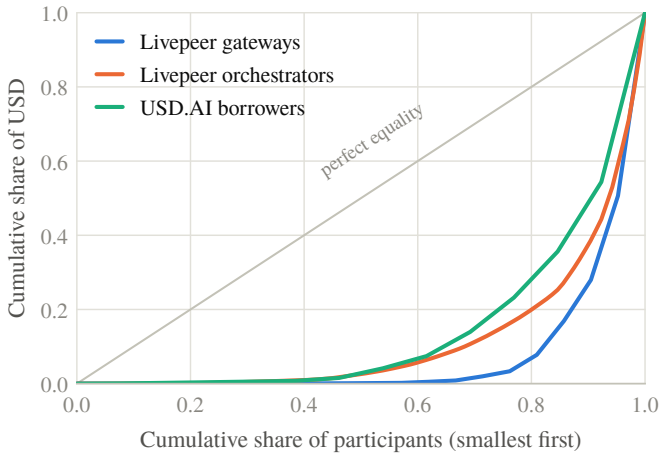


Fig. 19. Concentration of payers and payees: Livepeer gateways and orchestrators against USD.AI borrowers.

TABLE IX
GPU CREDIT AND DECENTRALIZED COMPUTE, STUDY WINDOW

Metric (study window)	USD.AI	Akash	Livepeer
USD volume	$290.1 M lent	$3.47 M spent	$1.01 M paid
Revenue to capital/providers	$6.94 M interest	$3.47 M	$1.01 M
Demand-side counterparties	13 borrowers	n/a	21 gateways
Supply-side counterparties	4 tranche lenders	58 providers	104 orchestrators
Top-1 share	45.6%	n/a	49.4%
Data source	GoldRush (Arbitrum)	Akash Console API	GoldRush (Arbitrum)

state at block 517,316,432 and a recovery rate on the GPU collateral. This is arithmetic, not a credit model; it ignores the time to repossess and sell hardware, legal costs and the fact that recoveries fall when every lender sells GPUs at once. The largest borrower holds 29.9% of outstanding principal and the three largest 70.5%. At a 40% recovery, a default by the largest borrower costs 5.9% of assets and leaves NAV at 1.054; the three largest cost 13.9% and push NAV to 0.965, below par. If every borrower defaults at the same recovery, NAV falls to 0.900. If the open escrow is also not returned, the loss is 42.9% of assets and NAV is 0.640.

Three features of the design decide how such a loss would play out. First, the vault prices deposits at an optimistic NAV and redemptions at a conservative one, and nothing in the contracts marks a loan down before a breach event. A borrower that stops paying would show up as a missed monthly repayment, then as a `LoanBreached` event, and only then in NAV. Holders who read the queue could leave at par before the mark. Second, exits are a queue. Today the idle deposit balance covers the pending queue 11 times over, so the vault can pay out without touching a loan. In a run the idle balance goes first, then the vault has to wait on 2.6 more years of loan maturities. Third, sUSDai is collateral elsewhere: 18.0% of the Ethereum supply is in Morpho Blue and 13.2% of the Arbitrum supply is in Pendle. A NAV mark below par would trigger liquidations in those venues that sell sUSDai into thin pools, pushing the market price further below NAV and feeding the queue.

The same mechanics apply to every protocol that lends

TABLE X
LOSS TO VAULT ASSETS UNDER DEFAULT SCENARIOS (ARITHMETIC ON THE RECONCILED STATE; RECOVERY IS ON GPU COLLATERAL)

Defaults	Recovery	Loss	Share of assets	NAV after
largest borrower	60%	$21.8 M	3.9%	1.076
three largest borrowers	60%	$51.4 M	9.3%	1.017
every borrower	60%	$72.9 M	13.1%	0.973
largest borrower	40%	$32.6 M	5.9%	1.054
three largest borrowers	40%	$77.1 M	13.9%	0.965
every borrower	40%	$109.3 M	19.7%	0.900
largest borrower	20%	$43.5 M	7.8%	1.033
three largest borrowers	20%	$102.8 M	18.5%	0.913
every borrower	20%	$145.8 M	26.3%	0.826
every borrower; escrow not returned	40%	$238.2 M	42.9%	0.640

stablecoin deposits against GPUs, including those that issue their own synthetic dollar [22]. An unwind would hit them together, because they share customers (AI labs renting compute), collateral (the same GPU generations), and the same secondary market for used hardware. Three questions separate the ones that would survive: how much of the yield comes from borrowers rather than from stablecoin incentives; how many borrowers there are and how large the biggest one is; and whether redemptions are backed by idle cash or by the hope of new deposits. All three can be measured from the chain with the code in this repository, and none of them appears in a headline APY.

C. Risks the chain shows only in outline

Seven risks are visible on-chain but cannot be sized from it.

Related parties. The same accounts submit the protocol's multisig transactions and the escrow facility's. The largest borrower, the fee recipient and the escrow facility all route PYUSD to the same off-ramp wallet. None of this is improper on its face, but a lender pricing "13 independent borrowers" needs the entity map behind those wallets, and the chain does not supply it.

Key compromise. The admin role can pause every core contract, and the escrow and multisig are controlled by a handful of externally owned accounts. Two days before this paper was finished, a leak of environment variables at a hosting provider let one attacker drain 231 wallets on 17 chains in about an hour, because the private keys sat on application servers [23]. The same failure mode applied to a signer here would not drain the vault directly, but it could pause redemptions or move escrow.

Fixed loans, floating benchmark. The loans are fixed at 7.0% to 15.0% for 2.6 more years. The yield holders compare against moves: Treasury bills, and the PYUSD and M^0 incentives that supplied most of the cash yield so far. If those fall, the vault's spread widens; if they rise, the vault holds below-market paper and the queue fills.

Custody, title and valuation. The hardware is described on-chain down to the server model and, for the early loans, the serial number (Section IV-B). What is not on-chain is who holds it, where, under what security filing or insurance, and whether the stated collateral values are appraisals or purchase

prices. At 66.8% book LTV on a three-year straight-line life, the margin for error in those values is small.

Bridged exits. 41.3% of shares live on Ethereum, Base, Plasma and Solana. A holder there must bridge back to Arbitrum before joining the queue. A paused bridge pauses the exit for two fifths of the supply.

Leverage on top. Pendle holds 13.2% of Arbitrum sUSDai and Morpho Blue 18.0% of Ethereum sUSDai. Fixed-yield buyers in Pendle have locked in today's rate; a NAV mark hits them first, and the leveraged positions sit there.

Classification. This is pooled private credit with a yield-bearing token. Payment-stablecoin rules in the United States now distinguish tokens that pay yield from those that do not, and the treatment of a structure like this one is not settled. A ruling could change who may hold it or how it may be marketed.

D. What this means for different readers

Traders. Three on-chain signals lead a NAV mark: a missed monthly repayment (today the longest gap is 25 days), a rise in the queue relative to idle cash (today $11\times$ covered), and the DEX discount of sUSDai to NAV. All three are computed by the repository and can be polled. The conservative redemption price (1.1172) against the deposit price (1.1203) is the protocol's own view of how much accrued interest is at risk.

Institutional allocators. Underwrite the escrow before the loans: 23.2% of assets is a claim with no collateral yet, most likely a refinancing in progress. Price the yield on borrower cash alone (28.4% coverage), not the headline. Treat the loan book as one neocloud exposure with 29.9% in a single name at 66.8% LTV on three-year-life hardware, and stress it against the recovery table (Table X). Ask who appraised the stated collateral values and what the fleet's utilization is; the chain gives the model and count but not the revenue.

Neoclouds. On-chain credit is cheaper to audit than bank credit, and USD.AI already publishes model, count and stated value for every pledge. A borrower that added utilization, rental revenue and its power contract would be the first a lender could underwrite from public data alone, and could reasonably ask for a lower rate than the 10.26% weighted average, which prices what is still unknown.

Researchers. The dataset is frozen and hash-verified, every number reproduces offline, and the method transfers to any GPU-credit protocol on an indexed chain. The open questions are the ones the chain cannot answer: independent collateral valuations, borrower revenue, and recoveries once the first loan defaults. A second protocol and a longer window would make this a comparative study.

E. Monitoring indicators

The pipeline can run on a schedule. We suggest six indicators, each computed by code in the repository:

1) *Cash coverage* κ by month, split into borrower interest and base yield.
2) *Pre-funded share*: `depositTimelockBalance` over `totalAssets`, with the age of each open escrow.
3) *Borrower HHI* on outstanding principal, and the largest borrower's share.
4) *Queue pressure*: pending redemptions against idle deposit balance, and the age of the oldest pending request.
5) *Missed payments*: loans whose last repayment is older than their repayment interval.
6) *Peg stress*: the market discount of USDai and sUSDai to NAV.

F. Other GPU-backed protocols

Other protocols also tokenize GPU financing or GPU revenue, and some issue their own synthetic dollars, such as GAIB's AID [22]. The same method works for any of them whose contracts run on a chain GoldRush indexes: decode from source, rebuild the loan book, reconcile against state and trace the cash. Comparing them on cash coverage, pre-funding and borrower concentration is a natural next study.

G. Data coverage

Without a unified API, several steps would have needed an archive node and custom indexing. GoldRush provided complete decoded histories for the lending contracts, event scans across 182 block windows, holder lists on four chains, holder lists at past blocks, one call listing every chain an address has used, cross-chain wallet balances and daily token prices. Three gaps remain. GoldRush does not offer `eth_call`, so the state check used a public RPC. Holder lists at past blocks are available only on its Foundational chains, so we have them for Ethereum but not Arbitrum. Akash and other Cosmos chains are not indexed.

H. Limitations

We study one protocol over 17 months. The loan book is rebuilt from events and depends on our decoder. We check it against vault state, but only at one block. Two topic hashes on the V1 router match no signature in the source history. Current holder lists give concentration at a single point in time. Contract labels come from GoldRush metadata and can be wrong. The GPU price series is a weekly median across clouds, not the price any borrower actually receives. Livepeer payments mix video transcoding and AI inference, and the Akash figures come from the network's own API. We make no claim about any off-chain entity. Wallets are described only by their on-chain role, and nothing here shows that funds were misused.

I. Ethics and right of reply

All data is public. We do not name individuals or link wallets to companies. We plan to share the escrow findings with USD.AI before publication and to include any response in a revised version.

VI. Conclusion

Chain data alone was enough to rebuild this GPU-backed dollar in detail. It shows \$290.1 M of GPU loans to 13 borrowers, a vault whose cash yield relies on stablecoin issuers, a redemption queue cleared in large monthly batches, and 23.2%

of assets held as pre-funded escrow instead of collateralized loans. It also shows that on-chain spending on compute is small next to on-chain lending for compute. Until GPU usage and revenue are recorded on-chain, lenders and curators have to estimate that part of the risk from off-chain information.

Disclosures

Conflicts of interest: The author has no financial relationship with USD.AI, GAIB, or any protocol, borrower or lender discussed in this paper, and holds no position in their tokens. **Use of AI:** An AI assistant (Claude) helped write the code and draft the text. The author reviewed the method and results, and the published code generates every number. **Data:** GoldRush API by Covalent; Akash Console API; GetDeploying GPU price snapshot (CC BY 4.0) [17]; Arbitrum and Solana public RPCs.

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